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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M.Tech (Electric Vehicle Technology) (CBCS) Degree Examinations

May - June 2024

22EVT222 – SOLAR BATTERY CHARGING SYSTEM

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain the terms insolation and irradiance with respect to solar cell.	6	L*2	1	3
b)	With the help of a equivalent structure of IGBT, explain its working principles.	7	L2	2	3
c)	Define Fill factor and efficiency of a solar cell. Hence find the efficiency of the PV module given below: A small PV module having an area of 0.094 m ² gives a current of 0.71 A and voltage of 16.5 V at maximum power point under STC. What is the maximum power point of the SPV module? Also, find out the efficiency.	7	L3	1	3
2. a)	What is a solar PV Module? Explain the I-V and P-V curves of a PV module.	6	L2	1	3
b)	With the help of a circuit diagram and waveforms, explain a Buck-Boost Converter operation in CCM mode.	7	L3	2	3
c)	In a PV power plant of 1 MW capacity, a large number of PV modules are connected in series. In such plants, 1 MW inverter can take input voltage in range of 400 V to 500 V. Design the number of PV modules to be connected in a single series (PV module string) to obtain voltage at maximum power point of 450 V. Also, estimate the peak power that will be supplied by one such PV module string. The parameters of PV modules to be used are: V _{oc} = 45 V, V _m = 35 V, I _{sc} = 8.2 A and I _m = 7.8 A	7	L3	2	3
3. a)	Explain the DC-AC 6 pulse inverters in 120° mode.	6	L2	2	3
b)	Why protective diodes are necessary in SPV system?	7	L3	1	3
c)	Enumerate the steps involved in the estimation of number of modules to be connected in parallel and their total power in designing a PV array.	7	L3	2	3
Unit – II					
4. a)	Discuss about the steps involved in wire size calculation in SPV system.	7	L2	4	3
b)	Explain the methods involved in estimating the number of batteries to be connected in series- parallel. Hence estimate the number of batteries to be connected in parallel to obtain 450 Ah capacity by using 12 V lead-acid batteries of 150 Ah. Also estimate total energy stored and power in array of batteries assuming rating hour as 10 hours.	13	L3	3	3
5. a)	Explain the terms SOC, DOD, C-rating with respect to Batteries.	7	L2	3	3
b)	Design a solar PV system for a house which contains 3 fans of 70 watts each running for 4 hours a day, 3 tubelights of 35 watts each running for 8 hours a day and a refrigerator of 250 watts running for 6 hours a day.	13	L4	4	3
6. a)	Explain the battery charge controllers.	7	L2	3	3
b)	Explain the methodology of designing and development of solar pump system.	13	L4	4	3
Unit – III					
7. a)	Write short notes on				
i)	Maintenance of solar PV systems.	10	L2	5	3
ii)	Maintenance of batteries in Solar PV systems.	10	L2	5	3
b)	Explain the components of a grid connected Solar PV systems.				
8. a)	Explain the safety precautions in installation of solar PV Systems.	10	L3	5	3
b)	Explain the steps in designing grid connected PV system for power plant.	10	L3	5	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
